

CNIDARIANS

This phylum includes jellyfish, corals, and anemones. They have stinging tentacles to catch live prey, which they digest in a simple sac-like gut.

All cnidarians are aquatic and most are marine. They have two body plans: a free-swimming, bell-like shape called a medusa – as seen in jellyfish – and a static polyp, typical of anemones. Neither medusae nor polyps have a head or front end. Their tentacles encircle a single gut opening, used to take in food and to eliminate waste.

A cnidarian's nervous system consists of a simple network of fibres; there is no brain. This means that the animal's behaviour is usually simple. Although they are carnivorous, cnidarians cannot actively pursue their prey, with the possible exception of the box jellyfish. Instead, most species wait for swimming animals to blunder into the reach of their tentacles.

STINGS

Cnidarians' outer skins – and in some species the inner skin as well – are peppered with tiny stinging capsules of a kind that is unique to animals of

this phylum. These stinging organs are called cnidocytes (or cnidocytes), and it is from these that cnidarians get their name. Cnidocytes are concentrated on the tentacles, and are triggered by touch or chemical signals, both when the animal comes into contact with potential prey and when it is under attack. Each cnidocyte contains a microscopic sac of venom and discharges a tiny coiled “harpoon” to inject the venom into flesh. Some stings can penetrate human skin and inflict severe pain, but the vast majority of cnidarians are harmless to humans.

ALTERNATING LIFE CYCLE

Many cnidarians have a life cycle that alternates between medusa and polyp – usually with one of these forms dominating, though in some groups one or other form is missing. The free-swimming medusa is normally the sexual stage. In most species fertilization is external: sperm and eggs are released in open water, forming planktonic planula larvae that resemble tiny flatworms. These settle in order to grow into polyps. Specialized polyps then produce new medusae to complete the cycle.

PHYLUM	CNIDARIA
CLASSES	6
ORDERS	22
FAMILIES	278
SPECIES	11,947



This microscopic view shows cnidocytes (stinging cells) that have been triggered to discharge their venom-laden harpoons.

BOX JELLYFISH

Members of the class Cubozoa are found in tropical and subtropical waters. They differ from true jellyfish in having a greater ability to control the direction and speed of their movement, rather than drifting with the currents.

A flapping skirt at the bottom of the bell powers them at considerable speed. Box jellyfish have eyes, set in clusters on the sides of their transparent bell. They can see enough to steer past obstacles and to spot prey.



0.3–3 m
1–10 ft

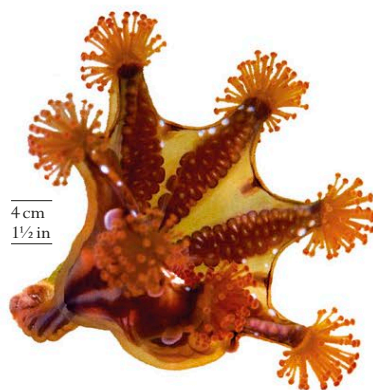
SEA WASP

Chironex fleckeri
F: Chirodropidae

This Indo-Pacific species – the largest box jellyfish – delivers a particularly painful sting and has caused human fatalities.

STALKED JELLYFISH

The adult is the only easily seen phase in the life cycle of members of the Class Staurozoa. Unlike other jellyfish it does not swim, but is attached to the substrate by a stalk. Eight arms radiate from the mouth, with clumps of tentacles at their tips. The other life stages are the extremely small planula larvae and polyps (called stauropolyps).



4 cm
1½ in

STALKED JELLYFISH

Halicystus auricula
F: Halicystidae

This species is found attached to algae and seagrasses in shallow cooler waters of the Atlantic and Pacific, where it catches food using stinging cells on the tentacles.

TRUE JELLYFISH

The familiar jellyfish bell is the medusa stage of the Scyphozoa life cycle. The polyp is reduced, or in some deep-sea forms, is missing. Jellyfish polyps undergo strobilation, a process that produces new tiny medusae by budding. Rhizostome jellyfish lack a fringe of tentacles around the bell.

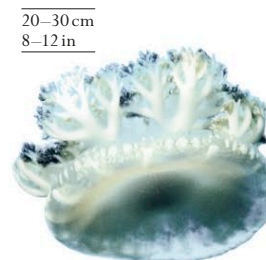


20–40 cm
8–16 in

MOON JELLYFISH

Aurelia aurita
F: Ulmaridae

This globally widespread genus has four long “arms” as well as small marginal tentacles. It swarms to breed near coasts and polyps settle in estuaries.



20–30 cm
8–12 in

SPOTTED LAGOON JELLYFISH

Mastigias papua
F: Mastigiidae

Like other rhizostome jellyfish, this algac-containing species traps planktonic food with mucus. It enters South Pacific lagoons, including those of oceanic islands.

UPSIDE-DOWN JELLYFISH

Cassiopea andromeda
F: Cassiopeidae

Superficially resembling an anemone, this Indo-Pacific rhizostome jellyfish lives at the bottom of lagoons, mouth upwards, with its bell pulsating to circulate water.

